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Inventor(s)	Shao; Qian et al.

Optical module

Abstract

An optical module includes a housing, a circuit board, a package, and at least one of a light-emitting assembly or a light receiving assembly. The package includes a package body and a soldering member. The package body includes a cavity. An end of the circuit board is inserted into the cavity, and the soldering member is located in a gap between the circuit board and the package body. The light-emitting assembly or the light receiving assembly is located in the cavity and electrically connected to the circuit board. The light-emitting assembly is configured to convert an electrical signal from the circuit board into an optical signal and emit the optical signal to an outside of the optical module, and the light receiving assembly is configured to convert the optical signal from the outside of the light module into an electric signal and transmit the electric signal to the circuit board.

Inventors: **Shao; Qian (Shandong, CN), Liu; Weiwei (Shandong, CN), Yao; Jianwei (Shandong, CN)**

Applicant: **HISENSE BROADBAND MULTIMEDIA TECHNOLOGIES CO., LTD.** (Shandong, CN)

Family ID: **1000008781815**

Assignee: **HISENSE BROADBAND MULTIMEDIA TECHNOLOGIES CO., LTD. (Qingdao, CN)**

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Primary Examiner: Bello; Agustin

Attorney, Agent or Firm: MH2 TECHNOLOGY LAW GROUP LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of International Patent Application No. PCT/CN2021/134352, filed on Nov. 30, 2021, which claims priorities to Chinese Patent Application No. 202110024351.6, filed on Jan. 8, 2021, and Chinese Patent Application No. 202110026033.3, filed on Jan. 8, 2021, which are incorporated herein by reference in their entireties.

TECHNICAL FIELD
(1) The present disclosure relates to the field of optical communication technologies, and in particular, to an optical module.
BACKGROUND
(2) With the development of cloud computing, mobile Internet, video, and other new business and application scenarios, the development and progress of optical communication technology has become more and more important. In optical communication technology, the optical module is a tool for achieving interconversion between an optical signal and an electrical signal and is one of the key components in optical communication devices. With the development of optical communication technology, it is required that the transmission rate of optical modules continues to increase.
SUMMARY
(3) An optical module is provided. The optical module includes a housing, a circuit board, a package, and at least one of a light-emitting assembly or a light receiving assembly. The circuit board is located in the housing. The circuit board includes a circuit board body, an upper surface, a lower surface, a first circuit, a second circuit, and an internal circuit. The lower surface is disposed opposite to the upper surface; the first circuit is disposed on the upper surface; the second circuit is disposed on the upper surface; and the internal circuit is disposed on a middle layer of the circuit board body. The package is located in the housing. The package includes a package body and a soldering member. The soldering member is located in a gap between the circuit board and the package body. The package body includes a cavity, a socket, an optical window opening, and an optical window. The first circuit is disposed on an end of the circuit board body inserted into a cavity. The second circuit is disposed on an end of the circuit board body opposite to the first circuit and located outside the package body. The socket is communicated with the cavity, and an end of the circuit board is inserted into the cavity through the socket. The optical window opening is disposed opposite to the socket. The optical window is located at the optical window opening. The light-emitting assembly or the light receiving assembly is located in the cavity and is electrically connected to the circuit board. The light-emitting assembly is configured to convert an electrical signal from the circuit board into an optical signal and emit the optical signal to an outside of the optical module, and the light receiving assembly is configured to convert the optical signal from the outside of the light module into an electric signal and transmit the electric signal to the circuit board. At least one of the light-emitting assembly or the light receiving assembly is electrically connected to the first circuit, and the first circuit is electrically connected to the second circuit through the internal circuit. At least one of the optical signal emitted by the light-emitting

assembly or the optical signal received by the light receiving assembly is transmitted through the optical window.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to describe technical solutions in the present disclosure more clearly, the accompanying drawings to be used in some embodiments of the present disclosure will be introduced briefly. However, the accompanying drawings to be described below are merely some embodiments of the present disclosure, and a person of ordinary skill in the art may obtain other drawings according to these drawings. In addition, the accompanying drawings to be described below may be regarded as schematic diagrams and are not limitations on an actual size of a product, an actual process of a method and an actual timing of a signal to which the embodiments of the present disclosure relate.

(2) FIG. **1** is a diagram showing a connection relationship of an optical communication terminal, in accordance with some embodiments;

(3) FIG. **2** is a structural diagram of an optical network terminal, in accordance with some embodiments;

(4) FIG. **3** is a structural diagram of an optical module, in accordance with some embodiments;

(5) FIG. **4** is an exploded view of an optical module, in accordance with some embodiments;

(6) FIG. **5** is a structural diagram of an airtight package of an optical module in the related art;

(7) FIG. **6** is an exploded view of an airtight package of an optical module in the related art;

(8) FIG. **7** is a structural diagram of a package in an optical module, in accordance with some embodiments;

(9) FIG. **8** is an exploded view of a package in an optical module, in accordance with some embodiments;

(10) FIG. **9** is a structural diagram of a soldering member in an optical module, in accordance with some embodiments;

(11) FIG. **10** is a structural diagram of a circuit board in an optical module, in accordance with some embodiments;

(12) FIG. **11** is a structural diagram of a circuit board in an optical module from another perspective, in accordance with some embodiments;

(13) FIG. **12** is a sectional view of a circuit board in an optical module, in accordance with some embodiments;

(14) FIG. **13** is another structural diagram of a circuit board in an optical module, in accordance with some embodiments;

(15) FIG. **14** is an assembly diagram of a package body and a circuit board in an optical module, in accordance with some embodiments;

(16) FIG. **15** is a partial assembly sectional view of a circuit board and a package body in an optical module, in accordance with some embodiments;

(17) FIG. **16** is a diagram showing a path where water vapor penetrates into a package from the inside of a circuit board in an optical module, in accordance with some embodiments; and

(18) FIG. **17** is a diagram showing an assembly process of a circuit board and a package body in an optical module, in accordance with some embodiments.

DETAILED DESCRIPTION

(19) The technical solutions in some embodiments of the present disclosure will be described clearly and completely with reference to the accompanying drawings; however, the described embodiments are merely some but not all embodiments of the present disclosure. All other embodiments obtained by a person of ordinary skill in the art based on the embodiments of the present disclosure shall be included in the protection scope of the present disclosure.

(20) Unless the context requires otherwise, throughout the specification and the claims, the term “comprise” and other forms thereof such as the third-person singular form “comprises” and the present participle form “comprising” are construed as an open and inclusive meaning, i.e., “including, but not limited to.” In the description of the specification, the terms such as “one embodiment,” “some embodiments,” “exemplary embodiments,” “example,” “specific example,” or “some examples” are intended to indicate that specific features, structures, materials, or characteristics related to the embodiment(s) or example(s) are included in at least one embodiment or example of the present disclosure. Schematic representations of the above terms do not necessarily refer to the same embodiment(s) or example(s). In addition, the specific features, structures, materials, or characteristics may be included in any one or more embodiments or examples in any suitable manner.
(21) Hereinafter, the terms such as “first” and “second” are used for descriptive purposes only and are not to be construed as indicating or implying the relative importance or implicitly indicating the number of indicated technical features. Thus, the features defined by “first” and “second” may explicitly or implicitly include one or more of the features. In the description of the embodiments of the present disclosure, the terms such as “a plurality of,” “the plurality of,” and “multiple” each mean two or more unless otherwise specified.
(22) In the description of some embodiments, the terms “coupled” and “connected” and their derivatives may be used. The term “connected” should be understood broadly. For example, the term “connected” may be a fixed connection, a detachable connection, or an integral connection; and it may be a direct connection or an indirect connection through an intermediate medium. However, the term “coupled” or “communicatively coupled” may also mean that two or more components are not in direct contact with each other but still cooperate or interact with each other. The embodiments disclosed herein are not necessarily limited to the content herein.
(23) The phrase “at least one of A, B, and C” has the same meaning as the phrase “at least one of A, B, or C,” and they both include the following combinations of A, B, and C: only A, only B, only C, a combination of A and B, a combination of A and C, a combination of B and C, and a combination of A, B, and C.
(24) The phrase “A and/or B” includes the following three combinations: only A, only B, and a combination of A and B.
(25) The use of the phrase “applicable to” or “configured to” herein means an open and inclusive expression, which does not exclude devices applicable to or configured to perform additional tasks or steps.
(26) The term such as “about,” “substantially,” or “approximately” as used herein includes a stated value and an average value within an acceptable range of deviation of a particular value. The acceptable range of deviation is determined by a person of ordinary skill in the art, considering measurement in question and errors associated with measurement of a particular quantity (i.e., limitations of a measurement system).
(27) Exemplary embodiments are described herein with reference to sectional views and/or plan views as idealized exemplary drawings. Therefore, variations in shapes with respect to the accompanying drawings due to, for example, manufacturing technologies and/or tolerances may be envisaged. Therefore, the exemplary embodiments should not be construed as being limited to the shapes of the regions shown herein, but including deviations due to, for example, ideally, an outline of an orthographic projection of a film pattern on a certain plane has a rectangular shape. Thus, the regions shown in the accompanying drawings are schematic in nature, and their shapes are not intended to show actual shapes of regions in a device and are not intended to limit the scope of the exemplary embodiments.
(28) In an optical communication system, an optical signal is used to carry information to be transmitted, and the optical signal carrying the information is transmitted to an information processing device such as a computer through an information transmission device such as an

optical fiber or an optical waveguide, so as to achieve transmission of the information. Since light has a characteristic of passive transmission when being transmitted through the optical fiber or the optical waveguide, low-cost and low-loss information transmission may be achieved. In addition, since a signal transmitted by the information transmission device such as the optical fiber or the optical waveguide is an optical signal, and a signal that may be recognized and processed by the information processing device such as a computer is an electrical signal, in order to establish information connection between the information transmission device such as the optical fiber or the optical waveguide and the information processing device such as the computer, there is a need to achieve interconversion between the electrical signal and the optical signal.
(29) In the field of optical communication technology, an optical module may achieve the interconversion between the optical signal and the electrical signal. The optical module includes an optical port and an electrical port. The optical module achieves optical communication with the information transmission device such as the optical fiber or the optical waveguide through the optical port and achieves electrical connection with an optical network terminal (e.g., an optical modem) through the electrical port. The electrical connection is mainly used for power supply, inter-integrated circuit (I2C) signal transmission, data information transmission, grounding, and the like. The optical network terminal transmits the electrical signal to the information processing device such as the computer through a network cable or wireless fidelity (Wi-Fi).
(30) FIG. **1** is a diagram showing a connection relationship of an optical communication system, in accordance with some embodiments. As shown in FIG. **1**, the optical communication system includes a remote server **1000**, a local information processing device **2000**, an optical network terminal **100**, an optical module **200**, an optical fiber **101**, and a network cable **103**.
(31) One end of the optical fiber **101** is connected to the remote server **1000**, and another end of the optical fiber **101** is connected to the optical network terminal **100** through the optical module **200**. An optical fiber itself may support long-distance signal transmission, such as signal transmission over several kilometers (6 kilometers to 8 kilometers). On this basis, if a repeater is used, in theory, infinite distance transmission may be realized. Therefore, in a typical optical communication system, a distance between the remote server **1000** and the optical network terminal **100** may typically reach several kilometers, tens of kilometers, or hundreds of kilometers.
(32) One end of the network cable **103** is connected to the local information processing device **2000**, and another end of the network cable **103** is connected to the optical network terminal **100**. The local information processing device **2000** includes one or more of a router, a switch, a computer, a mobile phone, a tablet computer, or a television.
(33) A physical distance between the remote server **1000** and the optical network terminal **100** is greater than a physical distance between the local information processing device **2000** and the optical network terminal **100**. Connection between the local information processing device **2000** and the remote server **1000** is accomplished by the optical fiber **101** and the network cable **103**, and connection between the optical fiber **101** and the network cable **103** is accomplished by the optical module **200** and the optical network terminal **100**.
(34) The optical module **200** includes an optical port and an electrical port. The optical port is configured to connect to the optical fiber **101**, so that bidirectional optical signal connection is established between the optical module **200** and the optical fiber **101**. The electrical port is configured to connect to the optical network terminal **100**, so that bidirectional electrical signal connection is established between the optical module **200** and the optical network terminal **100**. Interconversion between the optical signal and the electrical signal is achieved by the optical module **200**, so that information connection between the optical fiber **101** and the optical network terminal **100** is established. For example, an optical signal from the optical fiber **101** is converted into an electrical signal by the optical module **200** and then the electrical signal is input into the optical network terminal **100**, and an electrical signal from the optical network terminal **100** is converted into an optical signal by the optical module **200** and then the optical signal is input into

the optical fiber **101**. Since the optical module **200** is a tool for achieving the interconversion between the optical signal and the electrical signal and has no function of data processing, the information does not change in the photoelectric conversion process described above.
(35) The optical network terminal **100** includes a housing in a substantially cuboid shape, and an optical module interface **102** and a network cable interface **104** that are disposed on the housing. The optical module interface **102** is configured to connect to the optical module **200**, so that the bidirectional electrical signal connection between the optical network terminal **100** and the optical module **200** is established. The network cable interface **104** is configured to connect to the network cable **103**, so that bidirectional electrical signal connection between the optical network terminal **100** and the network cable **103** is established. Connection between the optical module **200** and the network cable **103** is established through the optical network terminal **100**. For example, the optical network terminal **100** transmits the electrical signal from the optical module **200** to the network cable **103** and transmits the electrical signal from the network cable **103** to the optical module **200**. Therefore, the optical network terminal **100**, as a master monitor of the optical module **200**, may monitor an operation of the optical module **200**. In addition to the optical network terminal **100**, the master monitor of the optical module **200** may further include an optical line terminal (OLT).
(36) A bidirectional signal transmission channel is established between the remote server **1000** and the local information processing device **2000** through the optical fiber **101**, the optical module **200**, the optical network terminal **100**, and the network cable **103**.
(37) FIG. **2** is a structural diagram of an optical network terminal. In order to clearly show a connection relationship between the optical module **200** and the optical network terminal **100**, FIG. **2** only shows structures of the optical network terminal **100** that are related to the optical module **200**. As shown in FIG. **2**, the optical network terminal **100** further includes a circuit board **105** disposed in the housing, a cage **106** disposed on a surface of the circuit board **105**, a heat sink **107** disposed on the cage **106**, and an electrical connector disposed inside the cage **106**. The electrical connector is configured to connect to the electrical port of the optical module **200**, and the heat sink **107** has protruding portions such as fins for increasing a heat dissipation area.
(38) The optical module **200** is inserted into the cage **106** of the optical network terminal **100**, and the optical module **200** is fixed by the cage **106**. Heat generated by the optical module **200** is conducted to the cage **106** and then is dissipated through the heat sink **107**. After the optical module **200** is inserted into the cage **106**, the electrical port of the optical module **200** is connected to the electrical connector inside the cage **106**, so that the bidirectional electrical signal connection between the optical module **200** and the optical network terminal **100** is established. In addition, the optical port of the optical module **200** is connected to the optical fiber **101**, so that the bidirectional optical signal connection between the optical module **200** and the external optical fiber **101** is established.
(39) FIG. **3** is a structural diagram of an optical module, in accordance with some embodiments. FIG. **4** is an exploded view of an optical module, in accordance with some embodiments. As shown in FIGS. **3** and **4**, the optical module **200** includes a shell and a circuit board **300**, a light-emitting assembly **400**, and a light receiving assembly **500** that are disposed inside the shell.
(40) The shell includes an upper shell **201** and a lower shell **202**, and the upper shell **201** covers the lower shell **202** to form the shell having two openings. An outer contour of the shell is generally in a cuboid shape.
(41) In some embodiments of the present disclosure, the lower shell **202** includes a bottom plate **2021** and two lower side plates **2022** that are located on two sides of the bottom plate **2021** and disposed perpendicular to the bottom plate **2021**. The upper shell **201** includes a cover plate **2011**, and the cover plate **2011** covers the two lower side plates **2022** of the lower shell **202** to form the shell.
(42) In some embodiments, the upper shell **201** includes a cover plate **2011** and two upper side plates that are located on two sides of the cover plate **2011** and disposed perpendicular to the cover

plate **2011**. The two upper side plates are combined with the two lower side plates, so that the upper shell **201** covers the lower shell **202**.
(43) A direction in which a connecting line between the two openings **204** and **205** extends may be the same as a length direction of the optical module **200** or may not be the same as the length direction of the optical module **200**. For example, the opening **204** is located at an end (a right end in FIG. **3**) of the optical module **200**, and the opening **205** is also located at an end (a left end in FIG. **3**) of the optical module **200**. Alternatively, the opening **204** is located at an end of the optical module **200**, and the opening **205** is located at a side of the optical module **200**. The opening **204** is the electrical port, and a connecting finger **301** of the circuit board **300** protrudes from the electrical port, and is inserted into the master monitor (e.g., the optical network terminal **100**). The opening **205** is the optical port and is configured to connect to an external optical fiber **101**, so that the optical fiber **101** is connected to the light-emitting assembly **400** and the light receiving assembly **500** inside the optical module **200**.
(44) By adopting an assembly mode of combining the upper shell **201** with the lower shell **202**, it is possible to facilitate installation of elements such as the circuit board **300**, the light-emitting assembly **400**, and the light receiving assembly **500** into the shell, and the upper shell **201** and the lower shell **202** may provide encapsulation and protection for these elements. In addition, in a case where elements such as the circuit board **300**, the light transmit-emitting assembly **400**, and the light receiving assembly **500** are assembled, it is possible to facilitate an arrangement of positioning components, heat dissipation components, and electromagnetic shielding components of these elements, which is conducive to implementation of automated production.
(45) In some embodiments, the upper shell **201** and the lower shell **202** are generally made of a metal material, which is conducive to electromagnetic shielding and heat dissipation.
(46) In some embodiments, the optical module **200** further includes an unlocking component **203** located outside the shell thereof, and the unlocking component **203** is configured to implement a fixed connection between the optical module **200** and the master monitor or to release a fixed connection between the optical module **200** and the master monitor.
(47) For example, the unlocking component **203** is located on outer sides of the two lower side plates **2022** of the lower shell **202** and has an engagement component that is matched with the cage of the master monitor (e.g., the cage **106** of the optical network terminal **100**). In a case where the optical module **200** is inserted into the cage of the master monitor, the engagement component of the unlocking component **203** fixes the optical module **200** in the cage of the master monitor. When the unlocking component **203** is pulled, the engagement component of the unlocking component **203** moves along with the unlocking component **203**, and then a connection relationship between the engagement component and the master monitor is changed to release the engagement relationship between the optical module **200** and the master monitor, so that the optical module **200** may be drawn out of the cage of the master monitor.
(48) The circuit board **300** includes circuit traces, electronic elements, and chips. Through the circuit traces, the electronic elements and chips are connected together according to a circuit design, so as to implement functions such as power supply, transmission of an electrical signal, and grounding. The electronic elements include, for example, a capacitor, a resistor, a triode, and a metal-oxide-semiconductor field-effect transistor (MOSFET). The chips include, for example, a microcontroller unit (MCU), a limiting amplifier, a clock and data recovery (CDR) chip, a power management chip, a digital signal processing (DSP) chip, or a transimpedance amplifier (TIA).
(49) The circuit board **300** is generally a rigid circuit board, and the rigid circuit board may also implement a support function due to its relatively hard material. For example, the rigid circuit board may stably support the electronic elements and the chips. The rigid circuit board may also be inserted into the electrical connector inside the cage **106** of the master monitor.
(50) The circuit board **300** further includes the connecting finger formed on an end surface thereof, and the connecting finger is composed of a plurality of independent pins. The circuit board **300** is

inserted into the cage **106** and is conductively connected to the electrical connector inside the cage **106** through the connecting finger. The connecting finger may be disposed on only one surface (e.g., an upper surface shown in FIG. **4**) of the circuit board **300** or may be disposed on both upper and lower surfaces of the circuit board **300** to adapt to an occasion where a large number of pins are needed. The connecting finger **301** is configured to establish electrical connection with the master monitor, so as to implement functions such as power supply, grounding, **120** signal transmission, or data signal transmission.
(51) Of course, flexible circuit boards are also used in some optical modules. A flexible circuit board is generally used in conjunction with a rigid circuit board as a supplement for the rigid circuit board.
(52) In order to improve the transmission rate of the optical module **200** and adapt to various harsh environments, the light-emitting assembly **400** and the light receiving assembly **500** of the optical module **200** need to be airtight. Therefore, the light-emitting assembly **400** and the light receiving assembly **500** are generally of a hermetic package structure.
(53) FIG. **5** is a structural diagram of an airtight package of an optical module in the related art. FIG. **6** is an exploded view of an airtight package of an optical module in the related art. As shown in FIGS. **5** and **6**, in the related art, an airtight package **600** of the optical module **200** mainly includes a metal cover plate **610**, a metal package **620**, and a ceramic circuit board **630**, and the metal package **620** includes a package opening **621** and a metal socket **622**. The metal cover plate **610** covers the package opening **621** located at an upper portion of the metal package **620**, the metal cover plate **610** and the metal package **620** form a sealed cavity, and optical elements such as the light-emitting assembly **400** or the light receiving assembly **500** are disposed in the sealed cavity. The metal cover plate **610** covers the package opening **621** of the metal package **620**, so as to form the sealed cavity through the metal cover plate **610** and the metal package **620**, thereby ensuring the sealing performance of the airtight package **600** to the optical elements. The metal socket **622** is disposed on a side of the metal package **620**, and the metal socket **622** is communicated with the sealed cavity; one end of the ceramic circuit board **630** is inserted into the metal package **620** through the metal socket **622** and is sealed and soldered with the metal package **620**. The ceramic circuit board **630** is provided with metal circuits, and the optical elements in the sealed cavity of the airtight package **600** are electrically interconnected with the circuit board **300** through the metal circuits on the ceramic circuit board **630**.
(54) For example, the ceramic circuit board **630** is inserted into the metal package **620** through the metal socket **622**, and then the optical elements are placed in the sealed cavity of the airtight package **600**; after the optical elements are electrically connected to the ceramic circuit board **630**, the metal cover plate **610** is encapsulated with the metal package **620** through a parallel sealing process, so as to realize the airtight encapsulation of the airtight package **600**. After the ceramic circuit board **630** and the metal package **620** are soldered together through a high-temperature sintering process, one end of the ceramic circuit board **630** located outside the metal package **620** needs to be connected to the circuit board **300** directly or through a flexible circuit board, and the optical elements in the airtight package **600** are driven to operate through the chips on the circuit board **300**.
(55) In the related art, the design manner to realize the encapsulation of the ceramic circuit board **630** and the metal package **620** is as follows: first, the ceramic green body is metallized; for example, pre-designed circuits are fabricated on the ceramic green body by using metal paste (e.g., high-melting-point metal heating resistor pastes of tungsten, molybdenum and manganese) through punching, hole filling, and printing through the high temperature co-fired ceramic (HTCC) process; then the ceramic circuit board **630** is finally manufactured by the processes of lamination, high-temperature sintering, and the like. Then, the ceramic circuit board **630** and the metal package **620** are soldered together through a high-temperature sintering process, so as to realize electrical interconnection inside and outside the airtight package **600** while maintaining the airtightness of

the airtight package **600**.

(56) However, the high-temperature co-fired ceramic process used in the ceramic circuit board **630** of the airtight package **600** in the related art is complex, technically difficult, and costly, and at present, there is no mature solution to realize the signal transmission rate above 10 Gb/s on the ceramic circuit board **630**. In order to reduce costs and improve high-speed signal quality, non-airtight packages have been introduced into the field of data centers. However, the non-airtight packages can only be used in the data center computer room with good environment and cannot meet the use in outdoor and other harsh environments, such as wireless 5G and other fields.

(57) In order to solve the above problems, in some embodiments of the present disclosure, a package **700** is provided. FIG. **7** is a structural diagram of a package in an optical module, in accordance with some embodiments. FIG. **8** is an exploded view of a package in an optical module, in accordance with some embodiments. As shown in FIGS. **7** and **8**, the package **700** in some embodiments of the present disclosure includes a package body **710** (e.g., a metal package) and a soldering member **720** (e.g., a solder member). A cavity is provided inside the package **700**. The cavity is, for example, a sealed cavity, and optical elements are encapsulated in the sealed cavity. The package body **710** includes a socket **711** and an optical window opening **712**. The socket **711** and the optical window opening **712** are disposed on two opposite sidewalls of the package body **710**, one end of the circuit board **300** is inserted into the package body **710** through the socket **711** of the package body **710**, and the optical window opening **712** is configured to realize optical communication with information transmission devices such as optical fibers or optical waveguides.

(58) In some embodiments of the present disclosure, the package body **710** further includes an optical window **713** mounted at the optical window opening **712**, so as to realize the sealing of the optical window opening **712** of the package body **710**, and the optical window **713** may be a glass sheet that allows light to pass through. An optical fiber adapter is connected to the optical window opening **712** of the package body **710** and transmits beams to the optical elements in the package **700** through the optical window **713**.

(59) In some embodiments of the present disclosure, in order to enhance the transmittance of the optical window **713** and prevent light reflection from affecting the performance of the optical elements in the sealed cavity of the package **700**, during installation, the optical window **713** is usually inclined at a preset angle (usually 8°) relative to a surface where the optical window opening **712** is located. For example, in a case where a normal direction of the optical window opening **712** is a horizontal direction, the surface where the optical window opening **712** is located is in a vertical direction, and in this case, an included angle between the optical window **713** and the vertical direction is the preset angle. Moreover, the surface of the optical window **713** is coated with an anti-reflection film corresponding to a wavelength of light emitted by the light-emitting assembly **400** or a wavelength of the light received by the light receiving assembly **500**, so as to enhance the transmittance of the light.

(60) The light-emitting assembly **400** includes optical elements such as a laser, a collimating lens, a converging lens, an optical isolator, an optical multiplexer, an optical fiber coupler, and a thermoelectric cooler. The position of the laser is close to the circuit board **300** inserted into the package body **710** and away from the optical window opening **712**, so as to facilitate the electrical connection between the laser and the circuit board **300** through wire bonding, thereby driving the laser to emit beams. The lens (including the collimating lens and/or the converging lens) is disposed between the laser and the optical window opening **712** and is close to the optical window opening **712** of the package body **710**. In this way, the beams emitted by the laser enter the lens, and a divergent beam is converted into a converging beam through the lens, and the converging beam is converged and coupled into the optical fiber adapter through the optical window **713** to realize light emission.

(61) The light receiving assembly **500** includes optical elements such as an optical splitter, a lens array, a reflective prism, and an optical collimator. The optical collimator is disposed close to the

optical window opening **712** and transmits the optical signal from the optical fiber adapter to the optical splitter, and the optical splitter demultiplexes a composite beam into a plurality of (e.g., 4 or 8) laser beams. The plurality of laser beams are converged to an optical receiver (e.g., the light receiver being a PIN diode or an avalanche diode) through the lens array, so as to realize light reception. The lens array is disposed close to the optical window opening **712** of the package body **710**. The lens array, the optical receiver, and the trans-impedance amplifier are sequentially disposed in the package body **710** along an optical path transmission direction, and the optical receiver and the trans-impedance amplifier are electrically connected to the circuit board **300** through wire bonding. In this way, the divergent beam transmitted by the optical fiber adapter is transmitted to the lens array through the optical window **713** and converted into a collimated beam through the lens array; the collimated beam enters the optical receiver, and the optical receiver converts the received optical signal into an electrical signal and outputs the electrical signal to the trans-impedance amplifier; and the electrical signal amplified by the trans-impedance amplifier is transmitted to the circuit board **300** to realize light reception.
(62) In some embodiments of the present disclosure, after the circuit board **300** is inserted into the package body **710**, a gap between the circuit board **300** and the package body **710** is sealed through the soldering member **720**. In order to facilitate the soldering of the package body **710** and the soldering member **720**, a surface of the socket **711** of the package body **710** has a metal layer, and the package body **710** and the soldering member **720** are directly soldered to each other. In some embodiments of the present disclosure, the metal layer on the surface of the socket **711** is made of metal that is easy to be soldered with the soldering member **720** (e.g., the solder member), and the metal layer is, for example, a gold-coated layer or a nickel-coated layer.
(63) FIG. **9** is a structural diagram of a soldering member in an optical module, in accordance with some embodiments. As shown in FIG. **9**, in some embodiments of the present disclosure, the soldering member **720** may be a structural member. The soldering member **720** includes a first side plate **721**, a second side plate **722**, and a third side plate **723**. The first side plate **721** is opposite to the third side plate **723**, and the first side plate **721** and the third side plate **723** are located on a side of the second side plate **722** close to the sealed cavity; and two ends of the second side plate **722** are connected to the first side plate **721** and the third side plate **723**. The second side plate **722** includes a through hole **7221** through which the circuit board **300** is inserted into the cavity of the package **700**.
(64) When the circuit board **300** is soldered to the package body **710** through the soldering member **720**, the first side plate **721** and the third side plate **723** of the soldering member **720** are soldered to positions of the package body **710** where the socket **711** is provided. After the circuit board **300** is inserted into the cavity of the package **700**, the circuit board **300** and the soldering member **720** are soldered to each other. In this way, the gap between the circuit board **300** and the package body **710** is completely sealed through the soldering member **720**, which makes the cavity of the package **700** the sealed cavity and the package **700** an airtight package.
(65) In some other embodiments of the present disclosure, the soldering member **720** includes a soldering material. After the circuit board **300** is inserted into the package **700** through the socket **711** of the package body **710**, the soldering material is applied to the gap between the circuit board **300** and the package body **710**, and the soldering material is soldered to the package body **710** and the circuit board **300**, and finally the gap between the circuit board **300** and the package body **710** is completely sealed.
(66) In some embodiments of the present disclosure, the soldering member **720** includes a solder paste, a solder sheet, or a solder wire.
(67) FIG. **10** is a structural diagram of a circuit board in an optical module, in accordance with some embodiments. FIG. **11** is a structural diagram of a circuit board in an optical module from another perspective, in accordance with some embodiments. As shown in FIGS. **10** and **11**, the circuit board **300** includes a circuit board body **300**A, an upper surface **301**, a lower surface **302**, a

first side **303**, a second side **304**, and a third side **305**. The upper surface **301** is disposed opposite to the lower surface **302**, and the upper surface **301** is disposed toward the upper shell **201** of the optical module **200**. The first side **303** is disposed opposite to the third side **305**, and two ends of the second side **304** are connected to the first side **303** and the third side **305**, respectively.

(68) In some embodiments of the present disclosure, the circuit board **300** includes a rigid multi-layer printed circuit board (PCB), and boards of the PCB are bonded together through an adhesive, and layers of the PCB are bonded together through an adhesive. The PCB is generally a copper clad laminate, and the copper clad laminate includes a substrate, a copper foil, and an adhesive. The substrate is an insulating laminate composed of polymer synthetic resin and reinforcing materials. The surface of the substrate is covered with a layer of pure copper foil with high conductivity and good weldability. A copper clad laminate with copper foil covered on one side of the substrate is called a single-sided copper clad laminate, and a copper clad laminate with copper foil covered on both sides of the substrate is called a double-sided copper clad laminate. The ceramic circuit board **630** in the related art is different from the circuit board **300** provided in some embodiments of the present disclosure.

(69) In order to facilitate blocking and soldering the gap between the circuit board **300** and the package body **710** through the soldering member **720**, the optical module **200** further includes a water vapor barrier layer **900** (as shown in FIG. **12**), and the water vapor barrier layer **900** is located at the connection between the circuit board **300** and the socket **711**. In some embodiments of the present disclosure, the water vapor barrier layer **900** includes a metal layer. For example, the metal layer may include a copper layer, and the surface of the copper layer is coated with a gold layer, and the circuit board **300** is soldered to the soldering member **720** through the water vapor barrier layer **900**. The soldering member **720** is located in the gap between the water vapor barrier layer **900** of the optical module **200** and the socket **711**, and the water vapor barrier layer **900** is soldered and fixed to the socket **711** through the soldering member **720**, so as to realize the hermetic encapsulation of the package body **710** and the circuit board **300**. In some embodiments of the present disclosure, the water vapor barrier layer **900** may be applied to portions of the circuit board **300** where the upper surface **301**, the lower surface **302**, the first side **303**, and the third side **305** are inserted into the package **700**, and the water vapor barrier layer **900** may be applied to portions of these surfaces where no portion of the package **700** is inserted that need to be soldered to the soldering member **720**, and may not be applied to the remaining portions. In this way, it is not only convenient for the upper surface **301**, the lower surface **302**, the first side **303**, and the third side **305** of the circuit board **300** to be soldered and fixed to the soldering member **720**, but it also prevents external water vapor from penetrating into the inside of the package **700** through the gap between the circuit board **300** and the package body **710**, which affects the performance of the optical elements disposed in the package **700**.

(70) Since the circuit board **300** is of a multi-layer board structure, the boards of the circuit board **300**, and the joints between layers of the circuit board **300** are easily penetrated by water vapor. In order to prevent water vapor from penetrating into the package **700** through the circuit board **300**, some embodiments of the present disclosure further provide a water vapor barrier layer **900** on the second side **304** of the circuit board **300**.

(71) FIG. **12** is a sectional view of a circuit board in an optical module, in accordance with some embodiments. FIG. **13** is another structural diagram of a circuit board in an optical module, in accordance with some embodiments. FIG. **14** is an assembly diagram of a package body and a circuit board in an optical module, in accordance with some embodiments. FIG. **15** is a partial assembly sectional view of a circuit board and a package body in an optical module, in accordance with some embodiments. As shown in FIGS. **10**, **12** and **15**, the circuit board **300** further includes a first circuit **310**, a second circuit **320**, and an internal circuit **350**. The first circuit **310** is disposed on an end of the circuit board body **300**A inserted into the package **700**, the second circuit **320** is disposed on an end of the circuit board body **300**A opposite to the first circuit **310** and located

outside the package **700**, and the internal circuit **350** is located on a middle layer of the circuit board body **300**A. One end of the internal circuit **350** is electrically connected to the first circuit **310**, and another end of the internal circuit **350** is electrically connected to the second circuit **320**.
(72) Since the upper surface **301** and the lower surface **302** of the circuit board **300** each are covered with the water vapor barrier layer **900**, it is impossible to arrange circuit traces. In order to realize the electrical connection, the optical elements in the package **700** are electrically connected to the first circuit **310**, and the first circuit **310** is electrically connected to the second circuit **320** through the internal circuit **350** of the circuit board **300**. In this way, the circuit traces may be led to the internal circuit **350** located on the middle layer of the circuit board **300**, and then the circuit traces may be led back to the surface of the circuit board **300**.
(73) In order to realize the electrical connection between the first circuit **310**, the second circuit **320** and the internal circuit **350** of the circuit board **300**, the circuit board **300** further includes a first via **330** and a second via **340**. The first via **330** is disposed between the first circuit **310** and the internal circuit **350**, and the first circuit **310** is electrically connected to the internal circuit **350** through the first via **330**. The second via **340** is disposed between the second circuit **320** and the internal circuit **350**, and the second circuit **320** is electrically connected to the internal circuit **350** through the second via **340**. In some embodiments of the present disclosure, both the first via **330** and the second via **340** are blind holes, and walls of the first via **330** and the second via **340** each are provided with a conductive material, so that the first via **330** is communicated with the first circuit **310** and the internal circuit **350**, and the second via **340** is communicated with the second circuit **320** and the internal circuit **350**.
(74) In some embodiments of the present disclosure, the first circuit **310** and the second circuit **320** each may include a circuit pad, and a plurality of circuit pads are sequentially disposed along a width direction of the circuit board **300**. In some embodiments, the width direction of the circuit board **300** is perpendicular to the normal direction of the optical window opening **712**. For example, as shown in FIG. **13**, the first circuit **310** may include a plurality of first circuit pads **311**, and the plurality of first circuit pads **311** are sequentially disposed along the width direction of the circuit board **300**. The second circuit **320** may include a plurality of second circuit pads **321**, and the plurality of second circuit pads **321** are sequentially disposed along the width direction of the circuit board **300**.
(75) As shown in FIGS. **12** to **14**, in some embodiments of the present disclosure, the water vapor barrier layer **900** includes an exposed region **910**, and the exposed region **910** is disposed at the end of the circuit board **300** inserted into the package body **710** and is located on the upper surface **301** of the circuit board body **300**A. In some embodiments, the exposed region **910** is formed by peeling off a part of the water vapor barrier layer **900**, however, the present disclosure is not limited thereto. A bottom surface of the exposed region **910** is the upper surface **301** of the circuit board **300**, and the upper surface of the exposed region **910** is the upper surface of the water vapor barrier layer **900**. The first circuit **310** is disposed on the bottom surface of the exposed region **910** of the water vapor barrier layer **900**.
(76) FIG. **16** is a diagram showing a path where water vapor penetrates into a package from the inside of a circuit board in an optical module, in accordance with some embodiments. As shown in FIG. **16**, due to the problem when the circuit board **300** and the soldering member **720** are soldered to each other, such as air bubbles and even cracks may be generated at the soldering portion of the two parts, which will lead to air leakage between the package body **710** and the circuit board **300** at the soldering portion, thus causing water vapor to penetrate into the inside of the package **700** from the bubbles and cracks at the soldering portion and affecting the performance of the optical elements in the package **700**.
(77) In order to solve the above problem, in some embodiments of the present disclosure, after the package body **710** and the circuit board **300** are soldered together through the soldering member **720**, a first waterproof layer **724** (referring to FIG. **9**) is coated on a side of the first side plate **721**,

the second side plate **722**, and the third side plate **723** of the soldering member **720** away from the cavity, and the first waterproof layer **724** covers the outer surface of the soldering member **720**. In this way, water vapor is prevented from penetrating into the inside of the package **700** from the bubbles or cracks of the soldering member **720**. In some embodiments of the present disclosure, the first waterproof layer **724** may include a glue layer, which can prevent water vapor from penetrating into the inside of the package **700** after the glue is cured.

(78) In some embodiments, external water vapor may also enter the gap between the layers of the circuit board **300** through the portion of the circuit board **300** that is not inserted into the package body **710** where the water vapor barrier layer **900** is not applied, and penetrate into the inside of the package **700** through the gap between the bottom surface of the exposed region **910** of the water vapor barrier layer **900** and the first circuit **310**, thereby affecting the performance of the optical elements inside the package **700**.

(79) In order to solve the above problems, in some embodiments of the present disclosure, after the first circuit **310** is electrically connected to the internal circuit **350** of the circuit board **300**, a second waterproof layer **313** (referring to FIG. **14**) is provided at the gap between the first circuit **310** and the bottom surface of the exposed region **910**. In this way, even if there is water vapor inside the circuit board **300**, the second waterproof layer **313** may block the water vapor, and the water vapor cannot enter the inside of the package **700** from the exposed region **910**.

(80) In some embodiments of the present disclosure, the second waterproof layer **313** may include a glue layer, and the bottom surface of the exposed region **910** is recessed toward the circuit board **300** relative to the water vapor barrier layer **900**, so that the glue will not flow around, and the glue will form a waterproof layer after curing, which may prevent water vapor from penetrating therethrough.

(81) FIG. **17** is a diagram showing an assembly process of a circuit board and a package body in an optical module, in accordance with some embodiments. As shown in FIGS. **13** to **15** and FIG. **17**, the water vapor barrier layer **900** is applied to portions of the circuit board **300** inserted into the package body **710** on each side of the circuit board **300**. The exposed region **910** is disposed at the end of the circuit board **300** inserted into the package body **710**, the first circuit **310** is disposed on the exposed region **910**, and the second waterproof layer **313** is provided at the gap between the first circuit **310** and the bottom surface of the exposed region **910**. The second circuit **320** is disposed at an end of the circuit board **300** that is not inserted into the package body **710**, and the internal circuit **350** is disposed at the middle layer of the circuit board **300**. The first via **330** is disposed between the first circuit **310** and the internal circuit **350**, and the second via **340** is disposed between the second circuit **320** and the internal circuit **350**. After cleaning the circuit board **300** and the package body **710** by means of plasma cleaning or ultrasonic cleaning, the circuit board **300** is inserted into the package body **710** through the socket **711**, so that the first circuit **310** is located in the cavity of the package body **710**, and the second circuit **320** is located outside the cavity of the package body **710**, and if necessary, glue may be used to pre-fix the junction of the two parts; then, the package body **710** and the circuit board **300** are soldered together through the soldering member **720**.

(82) After encapsulating the package **700**, the staff needs to perform visual inspection and leak detection to detect the assembly tightness of the circuit board **300** and the package body **710**, so as to prevent external water vapor from penetrating into the inside of the package **700** through the gap between the circuit board **300** and the package body **710**.

(83) A person skilled in the art will understand that the disclosed scope of the present disclosure is not limited to the specific embodiments described above, and some elements of the embodiments may be modified and replaced without departing from the spirit of the present disclosure. The scope of the present disclosure is limited by the appended claims.

Claims

1. An optical module, comprising: a housing; a circuit board located in the housing, and the circuit board including: a circuit board body; an upper surface; a lower surface disposed opposite to the upper surface; a first circuit, the first circuit being disposed on the upper surface; a second circuit, the second circuit being disposed on the upper surface; an internal circuit, the internal circuit being disposed on a middle layer of the circuit board body; a first via, the first via being disposed between the first circuit and the internal circuit; and a second via, the second via being disposed between the second circuit and the internal circuit; a package located in the housing, and the package including a package body and a soldering member, the soldering member being located in a gap between the circuit board and the package body; the package body including: a cavity, the first circuit being disposed on an end of the circuit board body inserted into a cavity; the second circuit being disposed on an end of the circuit board body opposite to the first circuit and located outside the package body; a socket, the socket being communicated with the cavity, and an end of the circuit board being inserted into the cavity through the socket; an optical window opening, the optical window opening being disposed opposite to the socket; and an optical window, the optical window being located at the optical window opening, and at least one of a light-emitting assembly or a light receiving assembly, the light-emitting assembly or the light receiving assembly being located in the cavity and being electrically connected to the circuit board, the light-emitting assembly being configured to convert an electrical signal from the circuit board into an optical signal and emit the optical signal to an outside of the optical module, the light receiving assembly being configured to convert the optical signal from the outside of the light module into an electric signal and transmit the electric signal to the circuit board; at least one of the light-emitting assembly or the light receiving assembly being electrically connected to the first circuit, and the first circuit being electrically connected to the second circuit through the internal circuit; wherein at least one of the optical signal emitted by the light-emitting assembly or the optical signal received by the light receiving assembly is transmitted through the optical window; an extension direction of the internal circuit is parallel to a normal direction of the optical window opening; and the first circuit is electrically connected to the internal circuit through the first via, and the second circuit is electrically connected to the internal circuit through the second via.
2. The optical module according to claim 1, wherein a surface of the socket connected to the soldering member has a metal layer.
3. The optical module according to claim 2, wherein the metal layer includes a gold-coated layer or a nickel-coated layer.
4. The optical module according to claim 1, wherein the optical window is inclined at a preset angle relative to a surface where the optical window opening is located.
5. The optical module according to claim 1, wherein the soldering member includes a first side plate, a second side plate, and a third side plate, the first side plate is disposed opposite to the third side plate, and two ends of the second side plate are connected to the first side plate and the third side plate, respectively; and wherein the second side plate includes a through hole, and the circuit board is inserted into the cavity through the through hole.
6. The optical module according to claim 5, wherein the first side plate and the third side plate are soldered with positions of the package body where the socket is provided, and the circuit board is soldered with the soldering member.
7. The optical module according to claim 1, wherein the soldering member includes a soldering material, and the soldering material is filled in the gap between the circuit board and the package body.
8. The optical module according to claim 7, wherein the soldering material includes a solder paste, a solder sheet, or a solder wire.

9. The optical module according to claim 1, wherein the circuit board further includes a first side and a third side opposite to each other, and the first side and the third side are located between the upper surface and the lower surface; and wherein the upper surface, the lower surface, and portions of the first side and the third side inserted into the cavity each have a water vapor barrier layer.
10. The optical module according to claim 9, wherein the water vapor barrier layer includes a metal layer.
11. The optical module according to claim 10, wherein the metal layer includes a copper layer, and a gold layer is coated on a surface of the copper layer.
12. The optical module according to claim 9, wherein a side of the soldering member away from the circuit board has a first waterproof layer.
13. The optical module according to claim 9, wherein the water vapor barrier layer includes an exposed region; wherein a bottom surface of the exposed region is the upper surface of the circuit board; and the first circuit is disposed on the bottom surface of the exposed region.
14. The optical module according to claim 9, wherein the circuit board further includes a second side, the second side is connected to the upper surface, the lower surface, the first side, and the third side, the second side is inserted into the cavity, and the second side has the water vapor barrier layer.
15. The optical module according to claim 14, wherein the water vapor barrier layer includes a metal layer.
16. The optical module according to claim 15, wherein the metal layer includes a copper layer, and a gold layer is coated on a surface of the copper layer.
17. The optical module according to claim 14, wherein a side of the soldering member away from the circuit board has a first waterproof layer.
18. The optical module according to claim 14, wherein the water vapor barrier layer includes an exposed region; and wherein a bottom surface of the exposed region is the upper surface of the circuit board; and the first circuit is disposed on the bottom surface of the exposed region.
19. The optical module according to claim 18, wherein a second waterproof layer is provided at a gap between the first circuit and the bottom surface of the exposed region.